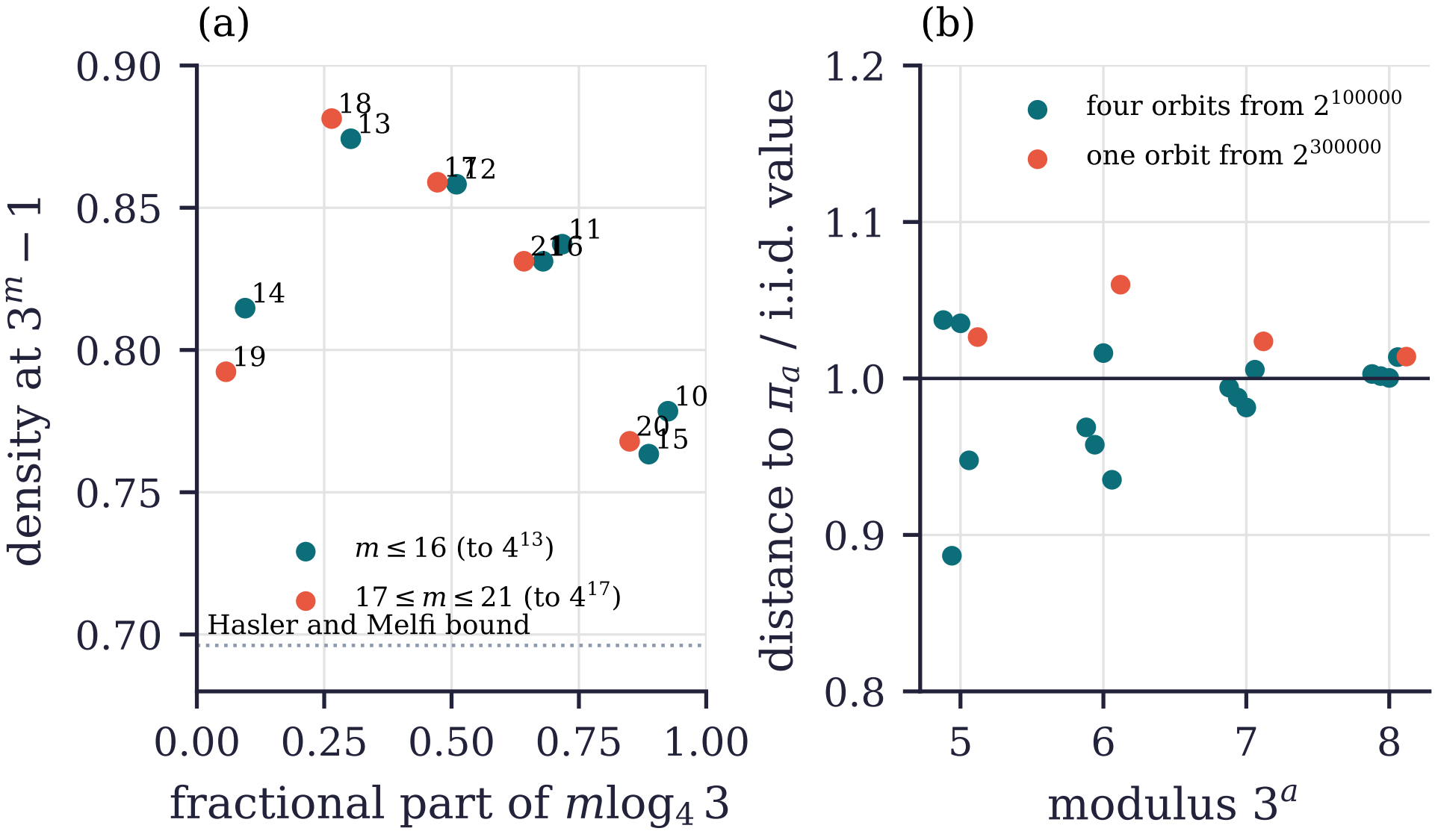
(a)
0.90
0.85
0.80
0.75
0.70
density at $3^m - 1$
18
13
17
12
11
16
14
19
10
20
15
$m \le 16$ (to 4^{13})
$17 \le m \le 21$ (to 4^{17})
Hasler and Melfi bound
0.00
0.25
0.50
0.75
1.00
fractional part of $m\log_4 3$
(b)
1.2
1.1
1.0
0.9
0.8
distance to π_a / i.i.d. value
four orbits from 2^{100000}
one orbit from 2^{300000}
5
6
7
8
modulus 3^a